SPOTTER ML ENGINEER ASSESSMENT

Freight Rate Prediction

Approach, results, and lessons learned

This is a forecasting problem

48,000

training loads

January–October 2025

12,000

validation loads

November–December 2025

The model predicts future months from past data. A random split does not represent that task.

Distance is the main pricing signal

Freight rates are strongly related to distance and market conditions.

A Ridge regression performed as well as more complex models in the first experiment.

Baseline Ridge

RMSE 633.9

The coordinates are redundant

0.9995

Correlation between haversine distance and the provided distance

The city coordinates are synthetic.

The provided distance was kept as the main geographic feature.

Validation follows time

Train through July
Test August

Train through Aug
Test September

Train through Sep
Test October

Target encoding and label screening used training rows only.

Log(rate) reduced percentage error

Approach	RMSE	MAPE
Baseline Ridge	633.9	8.4%
Ridge with log target	**633.0**	**5.8%**

The target transformation improved MAPE without materially changing RMSE.

Native categories gave the best RMSE

Model	RMSE
Native LightGBM	**628.7**
Three-model blend	629.0
LightGBM	629.9
CatBoost	630.1

Native categories beat the best blend by 0.3 RMSE.

Final model

Single model

Native-categorical LightGBM

RMSE	MAE	MAPE	R^2
628.7	113.8	5.11%	0.8265

Native categories produced the lowest out-of-fold error.

Two prediction files required two models

Validation predictions

12,000 rows

Used all columns and the selected native LightGBM model.

December predictions

31 daily rows

Used the six shared columns and a reduced CatBoost model.

The model matched the columns available for each output.